Spectral Anatomy of Legislative Networks: Methodology

1 Mathematical Framework

1.1 Notation and Preliminaries

Let $\mathcal{L}$ denote a legislature with m members voting on n distinct roll-call votes during a congressional session. We construct the **member-vote incidence matrix** $\mathbf{B} \in \{-1, 0, +1\}^{m \times n}$ where:

$$B_{ij} = \begin{cases} +1 & \text{if member } i \text{ votes Yea on vote } j \\ -1 & \text{if member } i \text{ votes Nay on vote } j \\ 0 & \text{if member } i \text{ is absent or abstains on vote } j \end{cases} \tag{1}$$

Without loss of generality, we assume $m \leq n$ (typically $m \approx 440$ for the House, $n \approx 500$–1000 votes per session).

1.2 Singular Value Decomposition

The spectral anatomy framework relies on the **Singular Value Decomposition** (SVD) of $\mathbf{B}$:

$$\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \sum_{k=1}^{r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top \tag{2}$$

where $r = \text{rank}(\mathbf{B}) \leq \min(m, n)$, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ are the singular values, and $\mathbf{U} \in \mathbb{R}^{m \times r}$, $\mathbf{V} \in \mathbb{R}^{n \times r}$ contain the left and right singular vectors respectively.

1.3 SVD Incidence Centrality

Following Franco et al. (2026), we define the **SVD Incidence Centrality** for members and votes as:

Definition 1 (Member Spectral Centrality)**.** The spectral centrality of member i is:

$$C_v(i) = \sum_{k=1}^{r} \frac{U_{ik}^2}{\sigma_k^2 + \epsilon} \tag{3}$$

where $\epsilon \geq 0$ is a regularization parameter (typically $\epsilon = 0$ with truncation at $k < r$ to avoid the null space).

Definition 2 (Vote Spectral Centrality)**.** The spectral centrality of vote j is:

$$C_e(j) = \sum_{k=1}^{r} \frac{V_{jk}^2}{\sigma_k^2} \tag{4}$$

2 Theoretical Justification

2.1 The $1/\sigma^2$ Weighting

The crucial innovation of SVD Incidence Centrality is the inverse-square weighting of singular values. This weighting scheme reflects the **spectral contribution** of each mode to the network topology:

Proposition 1 (Low-Frequency Emphasis)**.** The $1/\sigma_k^2$ weighting ensures that singular vectors corresponding to smaller singular values (near-zero modes) contribute disproportionately to centrality scores.

Interpretation: In the context of legislative networks:

- **Large singular values** ($\sigma \approx \sigma_1$): Capture high-variance patterns (party-line votes, popular bipartisan legislation)
- **Small singular values** ($\sigma \ll \sigma_1$): Capture low-frequency topological structure (Fiedler vector, near-consensus modes, cross-cutting cleavages)

The Fiedler vector (corresponding to the second smallest *non-zero* singular value, σ_{r-1}) specifically encodes the **bipartisan partition** of Congress. By emphasizing $1/\sigma^2$, we prioritize members and votes that define the *boundaries* of the legislative space rather than its *center of mass*.

2.2 Connection to Effective Resistance

SVD Incidence Centrality relates to effective resistance in the graph Laplacian framework. For the incidence matrix $\mathbf{B}$, consider the regularized Laplacian $\mathbf{L} = \mathbf{B}\mathbf{B}^\top + \epsilon\mathbf{I}$. The pseudoinverse $\mathbf{L}^\dagger$ relates to centrality via:

$$C_v(i) \approx [\mathbf{L}^\dagger]_{ii} \quad \text{as } \epsilon \to 0 \tag{5}$$

Note: The $1/\sigma^2$ weighting requires regularization ($\epsilon > 0$) to avoid divergence at $\sigma = 0$. In practice, we truncate at k components, which implicitly regularizes by excluding the null space.

This reveals that high-centrality members are those that, if removed, would most increase the "effective resistance" (structural rigidity) of the voting network.

3 Comparison to Existing Methods

3.1 DW-NOMINATE

DW-NOMINATE (Poole & Rosenthal) performs ideal point estimation via maximum likelihood:

- **Output:** Ideological position $x_i \in \mathbb{R}^d$ for each member
- **Assumption:** Voting is driven by spatial utility maximization
- **Limitation:** Captures ideology, not structural role

Key Difference: DW-NOMINATE asks "Where does this member sit on the left-right spectrum?" Spectral Anatomy asks "What is this member's contribution to the network's topological structure?"

Empirically, we expect weak correlation: a moderate member (center of DW-NOMINATE) may have low spectral centrality if they vote predictably with their party.

3.2 PageRank Centrality

PageRank models a random walk on the member similarity graph:

$$\pi = \alpha \mathbf{P} \pi + (1 - \alpha) \mathbf{v} \tag{6}$$

Key Difference: PageRank measures *popularity* (who votes similarly to many others). Spectral centrality measures *structural uniqueness* (who defines the network's eigen-directions).

3.3 Betweenness Centrality

Betweenness counts shortest paths passing through a node:

$$C_B(i) = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}} \tag{7}$$

Key Difference: Betweenness requires discrete "paths" in a binary graph. SVD Incidence Centrality operates directly on the incidence structure without thresholding, preserving continuous voting similarities.

4 Algorithmic Implementation

Algorithm 1 Compute Spectral Anatomy

Require: Roll-call vote matrix $\mathbf{B} \in \{-1, 0, +1\}^{m \times n}$, rank k
Ensure: Member centralities $C_v \in \mathbb{R}^m$, vote centralities $C_e \in \mathbb{R}^n$

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Filter: Remove members with < 50 votes, votes with < 10% participation
Center: B ← B − mean(B)        ▷ Optional: remove unanimous baseline
Compute truncated SVD: [U, Σ, V] = SVD(B, k)
Sort: Ensure σ_1 ≥ σ_2 ≥ ··· ≥ σ_k
Compute weights: w_k = 1/σ_k^2
Member centrality: C_v(i) = Σ_k U_ik^2 · w_k
Vote centrality: C_e(j) = Σ_k V_jk^2 · w_k
return C_v, C_e
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4.1 Computational Complexity

- **SVD computation:** $O(\min(mn^2, m^2n))$ for full decomposition
- **Truncated SVD (k components):** $O(mnk)$ using randomized methods
- **Memory:** $O(mn)$ for dense matrix, $O(\text{nnz})$ for sparse representation
- **Practical:** For $m = 440, n = 1000, k = 50$: < 1 second on standard hardware

5 Temporal Extension Methodology

To analyze the evolution of spectral structure across Congresses (1990–2024):

5.1 Sliding Window Analysis

For each Congress $t \in \{101, 102, \ldots, 118\}$:

1. Compute $\mathbf{B}^{(t)}$, $C_v^{(t)}$, $C_e^{(t)}$
2. Track member i's trajectory: $\{C_v^{(t)}(i)\}_t$ for members serving multiple terms
3. Identify **spectral regime changes**: sharp increases/decreases in centrality

5.2 Cross-Congress Stability

Compute the **spectral autocorrelation**:

$$\rho_t = \mathrm{corr}\left(C_v^{(t)}, C_v^{(t+1)}\right) \tag{8}$$

Low ρ_t indicates structural reorganization (e.g., after wave elections, party switches).

6 Hub/Authority Decomposition

We extend the framework to differentiate two structural roles:

6.1 Agenda Setters (Hubs)

Members who vote on disproportionately *consequential* votes (high-centrality votes):

$$H_i = \sum_j |B_{ij}| \cdot C_e(j) \tag{9}$$

6.2 Consensus Builders (Authorities)

Members whose voting patterns are *uniquely informative* about the network structure:

$$A_i = C_v(i) \cdot \mathrm{var}_j(B_{ij}) \tag{10}$$

High A_i indicates a member whose votes span multiple ideological dimensions (not predictable from party alone).

7 Robustness and Validation

7.1 Sensitivity Analysis

1. **Truncation rank** k: Vary $k \in \{10, 25, 50, 100\}$; verify centrality rankings are stable
2. **Vote threshold**: Vary minimum participation threshold (30, 50, 100 votes)
3. **Party switches**: Exclude members who switch parties mid-session

7.2 Null Model Comparison

Generate randomized null models:

1. **Permutation null**: Shuffle vote labels, preserving participation patterns

2. **Configuration model**: Preserve row/column marginals of $\mathbf{B}$
3. Compare observed centrality distribution to null distribution

7.3 External Validation

Correlate spectral centrality with:

- Committee assignments (leadership positions)
- Legislative effectiveness scores (Volden & Wiseman)
- Media coverage (GDELT database)
- Campaign contributions (OpenSecrets)

8 Interpretation Framework

8.1 What High Centrality Means

Unlike DW-NOMINATE's ideological center, spectral centrality identifies members who are **structurally load-bearing**:

- **Bridge members:** Vote across party lines on defining votes (e.g., Peterson)
- **Boundary definers:** Occupy extreme positions that define the spectrum (e.g., Amash, Massie)
- **Coalition kingmakers:** Vote with unique combinations of allies (e.g., Gaetz on McCarthy votes)

8.2 What High-Centrality Votes Mean

- **Party-defining votes:** Close partisan divisions that reveal party alignment
- **Coalition-fracturing votes:** Issues that split expected alliances
- **Outlier-revealing votes:** Lopsided votes where few dissenters have high leverage

9 Visualization Methodology

9.1 Spectral Embedding

Project members into 2D using the top two singular vectors:

$$\mathbf{x}_i = (U_{i1}, U_{i2}) \tag{11}$$

Color by party, size by $C_v(i)$.

9.2 Temporal Heatmaps

For each Congress, generate heatmaps of:

- Member centrality rankings (top 20)
- Vote centrality by policy area (using Voteview.com codes)
- Correlation between C_v and DW-NOMINATE first dimension

10 Novel Contributions

10.1 Counterfactual Spectral Analysis

We introduce **leave-one-out spectral analysis** to quantify each member's structural contribution to the legislative network. By systematically removing each member i and recomputing spectral properties, we define the **structural importance score**:

$$\mathcal{I}(i) = \frac{1}{3}\left(\frac{|\Delta\gamma|}{\gamma_0} + \frac{|\Delta\alpha|}{\alpha_0} + \frac{|\Delta r_{\text{eff}}|}{r_0}\right) \tag{12}$$

where $\Delta\gamma$ is the change in spectral gap, $\Delta\alpha$ is the change in algebraic connectivity, and Δr_{eff} is the change in effective rank.

This enables:

- **Predicting special election impacts**: Quantify how a member's departure affects network topology
- **Redistricting analysis**: Simulate the structural impact of replacing representatives
- **Strategic retirement analysis**: Identify when member departure would maximally disrupt the network

10.2 Spectral Coalition Prediction

We develop a method to predict coalition formation on upcoming votes using **spectral embeddings**. Each member i is represented by a k-dimensional embedding:

$$\mathbf{e}_i = \mathbf{u}_i \odot \sqrt{\mathbf{w}} \tag{13}$$

where $\mathbf{u}_i$ is the i-th row of $\mathbf{U}$ and $\mathbf{w}_k = 1/\sigma_k$ emphasizes structural modes.

For a bill with features $\mathbf{b}$ (policy area, partisan lean, salience), we predict coalition membership via cosine similarity in embedding space:

$$\mathbb{P}[\text{Yea}_i] = \frac{1 + \cos(\mathbf{e}_i, \mathbf{e}_{\text{bill}})}{2} \times \text{adjustment}(\text{party}_i, \mathbf{b}) \tag{14}$$

This is distinct from existing approaches:

- **IDEAL**: Predicts based on spatial utility (requires knowing bill's policy location)
- **Our method**: Predicts based on spectral similarity (captures latent coalition structures)

10.3 Network Role Taxonomy

Based on counterfactual analysis, we classify members into structural roles:

Definition 3 (Bridge Member)**.** A member where $\Delta\gamma > 0$ and $\Delta\alpha < 0$ upon removal. Their departure *increases* polarization.

Definition 4 (Polarizer)**.** A member where $\Delta\gamma < 0$ and $\Delta\alpha > 0$ upon removal. Their departure *decreases* polarization.

Definition 5 (Redundant Member)**.** A member where $|\Delta\gamma| < \epsilon$ and $|\Delta\alpha| < \epsilon$. The network is structurally invariant to their presence.

This taxonomy is **predictive**, not merely descriptive: bridge members are targets for lobbying, redundant members can be replaced without consequence.

11 Summary

The Spectral Anatomy framework provides a complementary lens to existing political science methods. By emphasizing low-frequency structural modes through $1/\sigma^2$ weighting, we identify the **load-bearing elements** of legislative networks—the members and votes that, if altered, would most reshape the topology of congressional politics.

Our **novel contributions**—counterfactual spectral analysis and spectral coalition prediction—transform the framework from descriptive to **predictive**. We can now:

1. Predict the structural impact of member turnover
2. Forecast coalition formation before votes occur
3. Identify optimal targets for legislative strategy

The methodology is computationally efficient ($O(mnk)$), robust to parameter choices, and extensible across time and institutional contexts. The central empirical question—whether load-bearing members are moderates or extremists—can now be addressed through counterfactual simulation across the 1990–2024 period.

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